

RESEARCH ARTICLE

Modulation of Antibody Glycosylation in Chinese Hamster Ovary Cells: Experimental and Computational Analyses Provide Mechanistic Insights

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ABSTRACT

Process conditions and cell culture media components significantly impact glycosylation in ways that are still poorly understood. Here, we evaluated the glycan distribution of CHO-K1 clones producing adalimumab, a biosimilar version of the anti-TNF antibody Humira. To modulate the glycan profile, the cell culture medium was supplemented with manganese and/or galactose. Both manganese and galactose increased galactosylation, and the effects were synergistic. Notably, the levels of Man5 were higher than in Humira, even when galactosylation was similar. qPCR analysis revealed few differences in glycosylation enzyme concentrations between different productivity clones or when 50 μ M manganese was added. We modeled the glycosylation pathways using a dynamic mathematical model to elucidate the mechanisms by which high mannose glycans increased and to develop a more predictive approach to culture modulation. Modeling results suggest that the concentration of uridine triphosphate (UTP), a component of activated sugars, is limiting and that increased uridine diphosphate (UDP)-galactose results in decreased UDP-N-acetylglucosamine, limiting complex glycan synthesis.

1 | Introduction

Biopharmaceuticals are complex molecules produced by living cells [1] using recombinant DNA methods, antibody technologies, or controlled gene expression. Biotherapeutics include hormones, growth factors, blood products, monoclonal antibodies (mAbs), fusion proteins, antibody-drug conjugates, and recombinant vaccines [2, 3].

mAbs are among the fastest-growing drug classes with an estimated market of ~\$230 billion in 2023 and a projected ~12% annual growth rate over the next decade ([4], <https://www.precedenceresearch.com/monoclonal-antibodies-market>). mAb

popularity stems from high specificity, low toxicity, and long circulation half-life [5–8], providing novel treatments for cancer, autoimmune, and inflammatory diseases, among others. Most approved therapeutic antibodies are manufactured in mammalian cell lines [7], with Chinese hamster ovary (CHO) cell lines serving as the preferred host [9] due to their extensive characterization, human-like glycosylation [10], and adaptability to suspension and serum-free culture [11, 12].

Protein glycosylation is the most common and well-characterized antibody post-translational modification, with N-linked glycosylation, involving the enzymatic addition of oligosaccharides to the amide nitrogen of asparagine residues, the most prevalent glycan

form [7, 13]. IgG antibodies contain a conserved N-glycosylation site at Asn297 within each CH2 domain of the Fc region [14, 15].

Patent expiration on many therapeutic proteins, particularly blockbuster mAbs, has provided opportunities for other companies to develop their own versions, known as “biosimilars” [16]. To gain approval, biosimilars must demonstrate high similarity to licensed reference products in physicochemical characteristics, efficacy, and safety, complying with the European Medicines Agency (EMA) and the US FDA regulatory guidelines [17, 18]. Glycosylation is a critical quality attribute (CQA) [19, 20], and controlling glycosylation remains a major challenge, as glycosylation variability may significantly impact clinical efficacy and safety [2, 21]. Moreover, achieving the desired glycan distribution often requires empirical optimization. Host cell selection, clonal variation, and bioprocess parameters, including medium composition, feeding strategy, culture duration, temperature, pH, and dissolved oxygen, influence glycosylation patterns [22].

As of 2024, the US FDA has approved ten biosimilars for Humira (adalimumab), AbbVie's blockbuster anti-tumor necrosis factor antibody, used to treat rheumatoid arthritis and other autoimmune disorders: Amjevita (Amgen), Cyltezo (Boehringer Ingelheim), Hadlima (Organon/Samsung Bioepis), Hulio (Biocon Biologics), Hyrimoz (Sandoz), Idacio (Fresenius Kabi), Yusimry (Coherus BioSciences, Inc.), Simlandi (Teva/Alvotect), Yuflyma (Celltrion), and Abrilada (Pfizer) [23]. Five of these biosimilars, Simlandi, Abrilada, Cyltezo, Yuflyma, and Hyrimoz, hold interchangeability designation, allowing substitution for Humira without prescriber approval. Most of these products are also EMA-approved, while at least seven biosimilars of Humira have been approved in China, most of which are unique entities [24]. Each manufacturer independently developed cell lines and bioprocesses to obtain regulatory approval, although several note minor glycan profile variations compared with the originator [25–30].

Although analytical characterization of mAbs is well established and documented, far less is publicly known about strategies to achieve biosimilarity as biopharmaceutical companies and CDMOs often keep this information proprietary. Edwards and colleagues reviewed glycosylation control strategies, highlighting persistent challenges with N- and O-linked glycosylation and predicting increasingly stringent regulatory expectations [31]. Both upstream (e.g., cell line engineering, fermentation bioprocess conditions, and media supplementation) and downstream (e.g., enzymatic remodeling and chromatography) approaches to match glycan patterns were described. Xu et al. applied a design-of-experiments (DOE) approach to match the glycan profile of a Herceptin biosimilar, noting that the conditions for optimal growth and productivity resulted in a highly dissimilar glycan profile [32]. Rameez and coworkers employed a multistage approach to modulating the high mannose glycans to control antibody-dependent cell-mediated cytotoxicity in producing a biosimilar of an undisclosed antibody [33]. They explored osmolarity, ammonium chloride, monensin, alternative feeds, and a variety of sugar and amino-sugar supplements in shake flasks, followed by a DOE study of mannose, glucosamine, and galactosamine. Finally, they performed a scale-up study in 3-liter bioreactors to optimize the sugars and amino sugars, feed strategy, and osmolarity. Recently, Gangwar and coworkers

explored various media additives using one-factor-at-a-time analysis, followed by a definitive screen design to match the glycosylation profile of a trastuzumab biosimilar using a statistical approach to match galactosylation, mannosylation, fucosylation, and sialylation indices [34]. This data-driven approach effectively modulated glycan profiles without significantly reducing titer or altering other CQAs.

Beyond biosimilarity efforts, several recent studies have investigated media additives that influence mAb glycosylation. Kranjc et al. evaluated enzyme inhibitors, ionophores, metal cofactors, sugar substrates, other substrates (e.g., N-acetylglucosamine, asparagine), and the toxic inhibitor, ammonium chloride, using a screening study followed by a DOE approach [35]. They assessed effects on high mannose glycans, afucosylation, terminal galactosylation, and terminal sialylation, as well as growth, viability, and titer. MnCl_2 (125 μM), galactose (79 mM), and monensin (100 nM) increased titers, whereas fucose (9.14 mM), 1-deoxynojirimycin hydrochloride (1 mM), and uridine (6.14 mM) reduced titers. Kifunensin (43 μM), monensin, arabinose (10 mM), and 1-deoxynojirimycin hydrochloride (1 mM) increased high mannose glycans. No significant reductions in high mannose glycans were obtained, but minor reductions occurred upon addition of NGI-1 (100 μM , an inhibitor of oligosaccharyltransferase), lithium chloride (10 mM), cytidine (6.2 mM), dexamethasone (1 μM), and copper (8 μM). Terminal galactosylation was increased by galactose, NGI-1, Ex-Cell GAL+, or fucose or lactose feeding in screening studies and increased by uridine and MnCl_2 in DOE studies. Similarly, Ehret and coworkers reported that 5–15 μM kifunensin increased high mannose glycans ~60%–80%, respectively [36]. Maximum galactosylation occurred with 48 μM Mn, 120 mM galactose, and 24 mM uridine, but a similar level was seen with 15 mM galactose in combination with manganese and uridine. Interestingly, the galactose, manganese, and uridine combinations doubled titers and significantly increased integrated viable cell densities (IVCD) relative to control.

In this study, we investigated adalimumab (Humira), a top-selling biopharmaceutical, focusing on glycan distributions across independently developed CHO cell clones and the impact of clonal variation. We compared gene expression patterns for glycosylation-related enzymes among clones and assessed whether media additives—manganese and galactose, individually or combined at literature-reported concentrations [37–39]—could align glycan profiles with the originator. While galactosylation was significantly enhanced, reducing high mannose glycans proved more difficult. To elucidate underlying mechanisms, we analyzed experimental data using a Golgi glycosylation model. Several different approaches have been described for glycosylation modeling, including modeling the Golgi as a series of stirred-tank reactors [40–43] or as a plug-flow reactor [44–47], glycosylation flux analysis [48–50], artificial neural networks [51–53], and dynamic Kriging [54]. We selected a cisternal maturation model that mechanistically describes how secretory pathway dynamics, glycosyltransferase availability, and activated sugar concentrations impact the synthesis and secretion of mAb Fc glycans, including high mannose species [45]. This framework is particularly suited to our dataset as it links cell-specific productivity and nucleotide sugar availability with N-glycan processing. Unlike data-driven [55] or fixed-cisternal models, which lack mechanistic resolution or constrain Golgi dynamics, the cisternal

maturation model captures key processing effects. Our analysis suggests that glycosyltransferase availability is not limiting and that depletion of uridine triphosphate (UTP), a precursor for biosynthesis of both uridine diphosphate-N-acetylglucosamine (UDP-GlcNAc) and uridine diphosphate-galactose (UDP-Gal), may underly high mannose glycan secretion.

2 | Materials and Methods

2.1 | Cell Line and Medium

CHO-K1 clones producing adalimumab were generated using PTSelect Technology as described previously [56]. Growth, productivity, and titer were assessed for several clones, and two clones (clone 9 and clone 14) with relatively high productivity were selected to examine glycosylation profiles. Cells were cultured in eCHO Basal Medium (Lonza), supplemented with 8 mM GlutaMAX (Thermo Fisher Scientific) and a 1:1000 dilution of anti-clumping agent (Thermo Fisher Scientific). Cells were routinely split every 3–4 days at a seeding density of 0.3×10^6 cells/mL in 125 mL shake flasks (Thomson Optimum Growth Flasks) with 40 mL working volume and were cultured in a humidified incubator using standard conditions of 37°C, 5% CO₂, and 120 rpm shaking speed.

2.2 | Shake Flask Cell Culture

All cultures for glycan analysis were inoculated at 0.3×10^6 cells/mL in 125 mL culture flasks (Thomson Optimum Growth Flasks) containing 40 mL of eCHO medium, supplemented with 8 mM GlutaMAX and a 1:1000 dilution of anti-clumping agent. Cultures were maintained in a humidified incubator at 37°C and 120 rpm with 5% CO₂. Feeding was started on Day 2 with daily feed additions of eCHO Feed Medium at 5% of the working volume. Supplemental glucose was added to maintain a concentration of 6 g/L. Cultures were harvested when viability dropped to ~60%.

Manganese (II) chloride (Mn) and D-(+)-galactose (Gal) were added at varying concentrations based on previous studies [37]. Stocks of Mn (40 mM) and Gal (500 mM) (Sigma-Aldrich) were prepared in eCHO basal and feed media. Cells were inoculated at 0.3×10^6 cells/mL in 125 mL shake flasks containing 40 mL of medium supplemented with Mn and/or Gal as shown in Table S1. Cultures were grown on orbital shakers at 120 rpm in a humidified 37°C incubator with 5% CO₂. Feeds containing Mn and/or Gal, as described in Table S1, were added to the culture at 5% v/v daily starting on Day 2. Glucose was measured daily and was maintained at 6 g/L. Samples were taken daily for cell counting, metabolite, and titer measurements. Media for antibody purification were harvested when cell viability dropped to approximately 60% and stored at –20°C. eCHO basal and feed media (Lonza) contain 7 and 37 nM MnSO₄•H₂O, respectively. The unsupplemented cultures exhibit cumulative manganese concentrations ranging from approximately 7 to 18 nM throughout the cell culture period, as shown in Figure S6.

2.3 | Cell Counts and Nutrient Analysis

Viable cell density and viability were measured daily by trypan blue dye exclusion using a Bio-Rad TC10 automated cell counter. Every 24 h, cell supernatants were collected, centrifuged at 14,500 rpm for 5 min, and stored at –20°C for subsequent analysis. Metabolites (glucose, lactate, glutamine, and glutamate) were measured by a YSI 2900 model biochemistry analyzer (YSI Life Sciences). Ammonium and potassium concentrations were measured using a 2974 ISE Ammonium Electrode and a 2975 ISE Potassium Electrode (YSI Life Sciences) according to the manufacturer's instructions. Total glutamine analysis was conducted by the addition of 1% v/v leucine peptidase (Sigma-Aldrich) to culture supernatants, followed by 2 h incubation at 37°C to cleave GlutaMAX as described previously [57].

2.4 | mAb Measurement

The concentration of recombinant human IgG in the culture supernatants was determined by Human IgG ELISA Antibody Pair Kit (STEMCELL Technologies) according to the manufacturer's protocol.

2.5 | mAb Purification

The culture supernatants containing recombinant adalimumab from the final day of fed-batch cultures were harvested and centrifuged at low and high speeds of $300 \times g$ for 15 min and $3000 \times g$ for 15 min, respectively. Then, 15 mL of supernatant was concentrated using Amicon Ultra-15 centrifugal filter units with a 30 kDa molecular weight cutoff (Millipore) to a final volume of ~1 mL. The concentrate was purified using a 1-mL NAb Protein A/G spin purification kit, (Thermo Fisher Scientific) according to the manufacturer's instructions. The purified IgG was subsequently concentrated with a 30 kDa cutoff Amicon Ultra 0.5 mL centrifugal filtration unit to a final volume of ~20 µL (Millipore) according to the manufacturer's instructions, and the concentration was measured by NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific) at 280 nm using an extinction coefficient of 14. Samples were frozen at –20°C for subsequent mAb product quality analysis.

2.6 | Sample Preparation for N-Glycan Profile Analysis

SDS-PAGE was performed under reducing conditions using 4%–20% Mini-PROTEAN TGX Precast Protein Gels (Bio-Rad) with Tris-glycine-SDS (TGS) running buffer (Sigma-Aldrich). The purified adalimumab samples were denatured with SDS sample buffer containing 91% 6× sample buffer and 9% 2-Mercaptoethanol (Bio-Rad) and then heated at 100°C for 3 min. 5 µg of each sample was loaded in each well. Proteins were separated at 200 V for ~35 min at room temperature. Gels were stained with Coomassie blue staining solution (0.1% w/v Coomassie blue, 40% v/v methanol, 10% v/v glacial acetic acid) at room temperature for 1 h or until the bands appeared. The gels were then destained in 50% v/v methanol and 10% v/v glacial acetic acid at room temperature until the background

was clear. Finally, the gels were soaked in water to remove the methanol and acetic acid. The heavy chain bands of each sample were cut out using razor blades, placed in microcentrifuge tubes containing sterile DI water, and sent for glycan analysis to the BOKU Mass Spectrometry Core Facility (Vienna, Austria). The originator Humira product (Humira; 40 mg/0.8 mL pre-filled syringe; AbbVie Inc., Chicago, IL, USA) was analyzed along with our samples.

2.7 | Glycopeptide Analysis

Glycopeptide analysis was performed at BOKU Core Facility Mass Spectrometry (CFMS), Vienna, Austria. An in-solution digestion of the samples was performed using 10–30 µg of protein. For the reduction of disulfide bridges, 15 mM dithiothreitol (DTT) dissolved in 0.1 M Tris/HCL buffer pH 7.8 was added to the sample, followed by incubation at 56°C for 45 min. Subsequently, carbamidomethylation of the reduced thiols was performed by adding 55 mM iodoacetamide (10 mg/mL) prepared in 0.1 M Tris/HCL buffer pH 7.8. The reaction was incubated for 30 min in the dark due to the light sensitivity of iodoacetamide.

Protein precipitation was carried out by adding four volumes of ice-cold acetone, followed by incubation at –20°C for 60 min. The precipitated proteins were recovered by centrifugation at 4°C at maximum speed for 5 min. The supernatant was carefully discarded, and the pellet was allowed to dry, using a speed vacuum concentrator.

For overnight digestion, the dried pellet was re-dissolved in 30 µL of 0.1 M ammonium bicarbonate buffer. Trypsin (Promega, sequencing grade, 0.1 µg/µL) was then added in a volume of 3–5 µL, and digestion was carried out at 37°C overnight. The digested samples were loaded on a nanoEase C18 column (nanoEase M/Z HSS T3 Column, 100Å, 1.8 µm, 300 µm × 150 mm, Waters) using 0.1% formic acid (FA) as the aqueous solvent. A gradient from 1% B (B: 80% Acetonitrile, 0.1% FA) to 40% B in 30 min was applied, followed by a 5 min gradient from 40% B to 95% B, which facilitates elution of large peptides, at a flow rate of 6 µL/min. Detection was performed with an Orbitrap MS (Exploris 480, Thermo) equipped with the standard H-ESI source in positive ion, DDA mode (= switching to MSMS mode for eluting peaks). MS scans were recorded (range: 350–2500 Da), and the eight highest peaks were selected for fragmentation. Instrument calibration was performed using Pierce FlexMix Calibration Solution (Thermo Scientific).

The possible glycopeptides were identified as sets of peaks consisting of the peptide moiety (EEQYNSTYR) and the attached N-glycan varying in the number of HexNAc (N) units, hexose (H), pentose (X), sialic acid (S), and deoxyhexose (F) residues. The theoretical masses of these glycopeptides were determined using the monoisotopic masses for amino acids and monosaccharides. Manual glycopeptide searches were made using FreeStyle 1.8 (Thermo). For the quantification of the different glycoforms, the peak intensities of the deconvoluted spectra were compared. Complete glycan profiles, including glycopeptide mass, are available as a supplementary Excel file.

2.8 | Quantitative Real-Time PCR (qRT-PCR)

mRNA expression levels of adalimumab heavy chain (HC), light chain (LC), and glycosylation enzymes were determined for fed-batch cultures of clones 9 and 14. Samples were cultured in biological duplicates as described in the shake flask cell culture section. Total RNA was extracted from approximately 1×10^7 cells on Day 6 and Day 9 using the RNeasy Mini Kit (Qiagen) or E.Z.N.A. Total RNA Kit I (Omega Bio-tek) according to the manufacturer's protocol. RNA concentration and purity were determined using a NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific) at 260 and 280 nm wavelengths. The cDNA was synthesized using 200 ng of total RNA and RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific) or First Strand cDNA Synthesis (Standard Protocol) (NEB #M0253) according to the manufacturer's instructions. All primers (Table S2) were designed using PrimerQuest Tool and synthesized by Integrated DNA Technologies, Inc. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as the housekeeping gene. qRT-PCR was performed with 40 cycles of amplification using a StepOnePlus Real-Time PCR System and PowerTrack SYBR Green Master Mix (Applied Biosystems) or CFX Opus 96 Real-Time PCR System (Bio-Rad) and Luna Universal qPCR Master Mix (NEB). All samples were analyzed in duplicate (technical replicates) for each target gene and the control gene GAPDH. The relative changes in gene expression of the IgG HC and LC throughout the culture, as well as for the cultures supplemented with Mn, were compared with the mRNA expression levels of clone 9 on Day 6 and the control on Day 6, respectively, using the $2^{-\Delta\Delta CT}$ method [58]. In this study, we observed variations in GAPDH expression between clones 9 and 14, which could create artifacts even though the variations were not statistically significant. Therefore, the threshold cycle of each gene was not normalized by GAPDH. Since GAPDH expression levels in cultures supplemented with Mn remained stable, normalizing the data using GAPDH would not provide meaningful insight. To maintain consistency, we avoided normalizing the data to GAPDH.

2.9 | Statistical Analysis

All statistical analyses were performed using GraphPad Prism version 8.0.1 for Windows, GraphPad Software, San Diego, CA, USA, www.graphpad.com.

3 | Results

3.1 | Comparison of Fed-Batch Shaker Flask Cultures of Clones 9 and 14 Shows Differences in Growth, Productivity, Gene Expression, and Glycan Profiles

3.1.1 | Cell Growth, mAb Productivity, and HC and LC mRNA Levels

Fed-batch shake-flask studies evaluated growth, specific productivity, product titer, and mRNA levels of clones 9 and 14 (Figure 1, Table 1). Clone 9 reached a higher maximum viable cell density (VCD) and IVCD than clone 14 and exhibited a 1.3-fold greater specific productivity (Qp) ($p = 0.0377$, unpaired *t*-test) and 1.3-fold

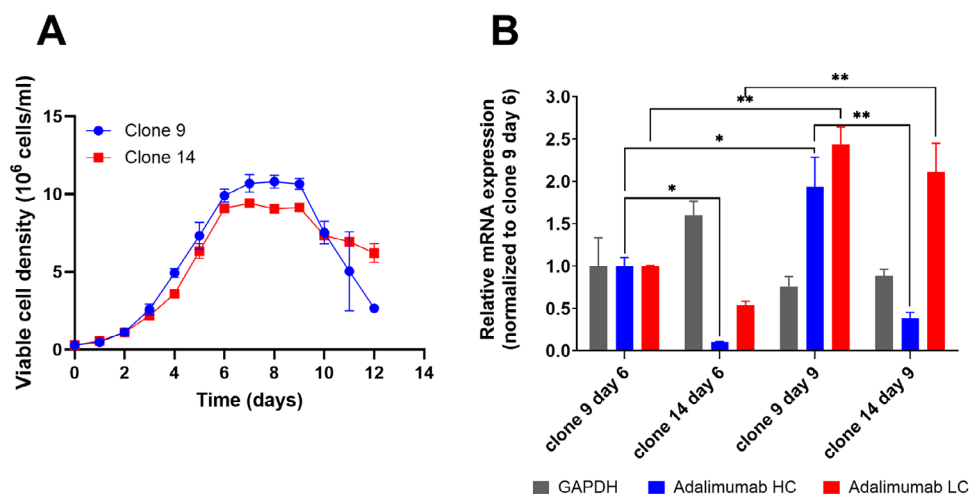


FIGURE 1 | Comparison of (A) viable cell densities (VCD) and (B) heavy chain and light chain mRNA levels on Days 6 and 9. The relative levels compared to the amount of heavy chain or light chain mRNA on Day 6 of clone 9 are shown. Error bars represent the standard deviation of duplicate biological samples. Data in 1B were analyzed by two-way ANOVA. * $p < 0.05$, ** $p < 0.01$.

TABLE 1 | Comparison of growth and productivity for fed-batch shake-flask cultures of clone 9 and clone 14.

Clone	Peak IVCD (10^6 cells/mL-days)	Qp (pg/cell/day)	Doubling Time (h)	Titer (mg/L)
Clone 9	72.0 ± 3.3	0.12 ± 0.000	23.4 ± 1.4	8.6 ± 0.9
Clone 14	68.1 ± 0.6	0.095 ± 0.007	28.2 ± 2.6	6.4 ± 0.8

higher titer than clone 14 (Figure 1A, Table 1), although the titer difference was not statistically different ($p = 0.1291$, unpaired t -test). HC mRNA levels were significantly higher in clone 9 than in clone 14 on Days 6 ($p = 0.0283$, two-way ANOVA) and 9 ($p = 0.0039$, two-way ANOVA), peaking on Day 9 ($p = 0.0247$, two-way ANOVA) (Figure 1B). Clone 9 showed a modest increase in LC mRNA relative to clone 14, peaking on Day 9 (Figure 1B). LC mRNA levels did not differ significantly between clones, but expression increased between Days 6 and 9, peaking on Day 9. Notably, the clones differed in growth and productivity under identical culture conditions, with productivity likely influenced by mRNA levels.

Glucose and lactate concentrations were monitored throughout cultivation (Figure S1). Both cultures had a final glucose concentration of ~ 4 g/L, indicating no glucose depletion. Neither clone showed a shift from lactate production to consumption, with lactate accumulating until termination.

3.1.2 | Glycan Analysis

To assess glycan clonal variation, we analyzed the N-glycosylation profiles of adalimumab purified from clones 9 and 14 (Figure 2) and compared them to profiles measured from the originator Humira and reported in the literature [17]. In both clones, CHO cells primarily expressed G0F glycans, followed by Man5 and then -NAF. (Nomenclature follows Peng and coworkers [59]). -NAF corresponds to G0F missing one N-acetylglucosamine, that is, a hybrid glycan. For less common structures, H = hexose, generally mannose; N = N-acetylglucosamine (GlcNAc); F = fucose. For

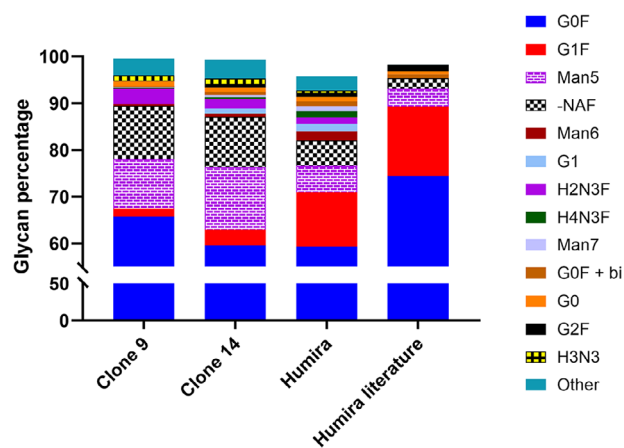


FIGURE 2 | Glycosylation profiles determined by glycopeptide mass spectrometry of adalimumab produced in two CHO cell lines, clones 9 and 14 in fed-batch shaker flasks compared with glycan profiles of originator Humira and values reported in the literature [17]. Glycans comprising less than 1% of all samples are excluded from the figure.

example, H2N3F contains two hexoses, three GlcNAcs, and one fucose; glycan structures are shown in Scheme S1 in the order listed in Figure 2. Clone 9 had higher G0F levels (66%) than clone 14 (60%), but lower G1F (2% vs. 3%) and Man5 (11% vs. 13%). Clone 9 also had slightly higher levels of -NAF (11%) and H2N3F (3%) than clone 14 (11% -NAF and 2% H2N3F). The levels of G0 and H3N3 were comparable between clones. Both clones produced $<1\%$ non-glycosylated glycans, whereas Humira contained 4%. As shown in Figure 2, glycosylation differs among

clones expressing the same antibody from the same host, likely reflecting natural glycan heterogeneity and the dynamic activity of hundreds of glycosylation enzymes in CHO cells [60]. G0F levels in the originator Humira (59%) were lower than reported in the literature (75%), while G1F and G2F were slightly reduced. Man5, -NAF, G0F+bi, and G0 were modestly higher than literature values. Glycans from a third clone (clone 16) contained a higher fraction of complex species (G0F 74%, G1F 14%, -NAF 4%, Man5 3%, G2F 1.6%; data not shown) than clones 9 or 14, but productivity was markedly lower, and this clone was not pursued further.

To determine whether altered glycosylation-related enzyme expression contributed to glycan variation, we quantified mRNA levels of 31 N-glycosylation-related genes, including glycosyltransferases and nucleotide sugar donor transporters in clones 9 and 14 on Days 6 and 9 of fed-batch cultures using qRT-PCR (Figure S2). Six genes (Man1a2, GLUT-1, Mgat4a, Mgat4b, Mgat5, and ST3Gal4) were differentially expressed between clones 9 and 14. Expression differences were observed on both Days 6 and 9, except for GLUT-1 and Mgat5, which differed only on Day 9. In all cases, transcript levels were higher in clone 14 than in clone 9. Mgat4a showed the largest fold difference, though its high cycle thresholds indicated low absolute expression. B4GalT2 expression did not differ between clones, but its profile shifted significantly between Days 6 and 9, peaking on Day 6.

Despite these differences, gene expression changes were uncorrelated with glycan variation. For example, N-acetylglucosaminyltransferases (Mgat1-5) transfer GlcNAc from the donor substrate UDP-GlcNAc (brought into the Golgi via transporter Slc35a3) to exposed mannose residues on glycoproteins, with Mgat1 and Mgat2 essential for G0F synthesis. Mgat1, Mgat2, UDP-GlcNAc pyrophosphorylase 1 (Uap1), and Slc35a3 expression were similar between clones on both days, yet G0F levels were higher in clone 9, indicating that enzyme expression alone does not account for the observed glycan differences.

Similarly, B4GalT1-3 enzymes add galactose (brought into the Golgi by the UDP-Gal transporter, Slc35a2) to terminal GlcNAc. Although clone 14 had slightly higher G1F levels than clone 9, B4GalT1-3 and Slc35a2 expression were comparable. Both clones also exhibited high Man5 levels, slightly higher in clone 14. Mannosidases (Man1a1, Man1a2, Man1c1, Man2a1, Man2a2), responsible for α -1,2-mannose trimming, were expressed at similar levels (Ct < 30, Figure S2A) in both clones, suggesting Man5 abundance was not controlled by these enzymes.

3.2 | Addition of Mn and/or Gal Into Fed-Batch Shaker Flask Cultures Alters Glycosylation Patterns

To modulate glycan profiles toward those of commercial Humira, we supplemented fed-batch shaker-flask cultures of clones 9 and 14 with varying concentrations of Mn (a cofactor for glycosyltransferases) and/or Gal (Table S1) and monitored effects on growth, productivity, and glycosylation.

3.2.1 | Cell Growth, Metabolism, and Antibody Production

In clone 9, supplementation with 0–60 mM Gal and/or 0–75 μ M Mn caused slight growth inhibition, with the 60 mM Gal alone condition showing the greatest reduction in VCD and IVCD (Figure 3A, Table 2). Specific productivity was largely unchanged across conditions, though a minor decrease in titer was observed compared to the control (Table 2).

In clone 14, Mn alone (0–75 μ M) did not affect maximum VCD or IVCD (Figures 3B, Table 2). Gal supplementation alone, or combined Mn + Gal, reduced VCD and IVCD in a dose-dependent manner, with 60 mM Gal alone causing the largest reduction. Specific productivity was similar to control in most conditions, though 30- and 60-mM Gal alone slightly increased specific productivity. Harvest titers were comparable across all conditions (Table 1).

Overall, Mn and/or Gal supplementation slightly slowed growth in both clones, with 60 mM Gal alone producing the lowest VCD and IVCD, particularly in clone 14 (Figure 3B, Table 2). Clone 14 had lower productivity and titer than clone 9, but experienced smaller decreases in response to supplementation. These effects likely reflect physiological changes induced by medium alterations [61].

Figure S3 shows glucose, lactate, glutamine, glutamate, and ammonium levels during cultivation. Mn and/or Gal supplemented cultures showed slightly increased glucose, likely reflecting slower growth. Lactate was not consumed under any condition, and ammonia accumulated until termination. Free glutamine was initially low, increased transiently, then sharply declined around Day 6 to near depletion, coinciding with reduced growth. The transient glutamine increase paralleled a rise in viable cell density, possibly providing additional aminopeptidases [57]. Glutamate rose throughout culture, though more slowly once glutamine was depleted, suggesting significant generation of glutamate from glutamine catabolism.

3.2.2 | Glycan Analysis

Mn and/or Gal supplementation significantly impacted adalimumab glycan profiles in clone 9, primarily reducing G0F glycans (Figure 4A, Table 3). A synergistic effect was observed with 50 μ M Mn + 30 mM Gal, decreasing G0F by 33%, 12% more than the sum of individual effects, and increasing G1F by 24%, 4% above the combined individual increases. An additive effect was seen with 25 μ M Mn + 15 mM Gal, yielding a 22% reduction in G0F glycans, consistent with the sum of individual decreases (4% Mn, 18% Gal). Supplementation with 0–60 mM Gal and/or 0–75 μ M Mn increased G1F and decreased -NAF glycans. The largest G1F increase (24%) occurred with 50 μ M Mn + 30 mM Gal. Man5 levels increased notably with combined Mn and Gal at 25 μ M + 15 mM and 50 μ M + 30 mM. In clone 9, G2F glycans increased synergistically with 50 μ M Mn + 30 mM Gal and additively with 25 μ M Mn + 15 mM Gal. G0F, G1F, -NAF, G1, and H2N3F levels with 50 μ M Mn alone approximated originator Humira, though Man5 and G2F were slightly elevated

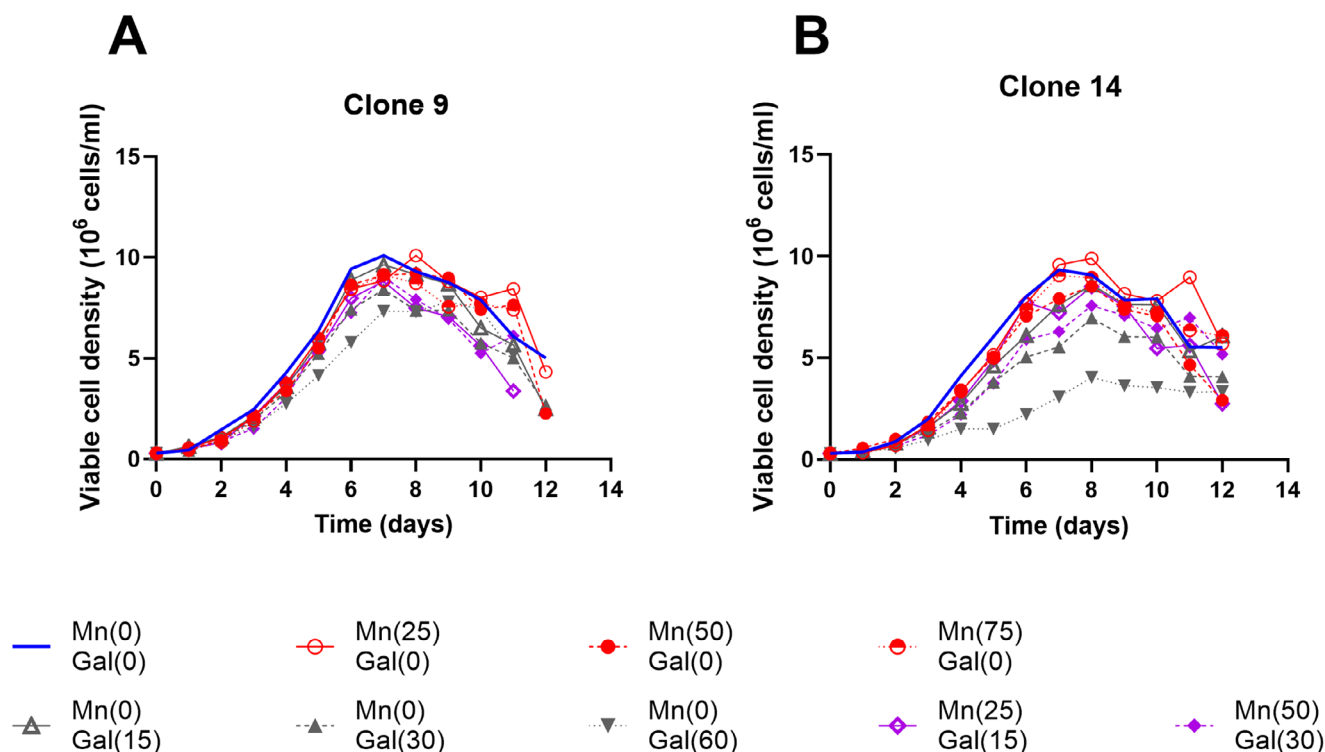


FIGURE 3 | Effects of manganese and/or galactose addition on VCD for clone 9 (A) and clone 14 (B) in fed-batch shake flasks.

TABLE 2 | Impact of manganese and galactose addition on growth and productivity in fed-batch shake flask cultures of clone 9 and clone 14.

Culture condition	Peak IVCD (10^6 cells/mL-days)		Qp (pg/cell/day)		Titer (mg/L)	
	Clone 9	Clone 14	Clone 9	Clone 14	Clone 9	Clone 14
Mn(0 ^a) Gal(0)	69.3	64	0.17	0.07	11.7	4.4
Mn(25) Gal(0)	68.4	66.3	0.11	0.06	7.8	3.9
Mn(50) Gal(0)	64.8	56	0.13	0.06	8.3	3.2
Mn(75) Gal(0)	62	60.8	0.11	0.06	6.3	3.5
Mn(0) Gal(15)	63.2	56.3	0.12	0.06	7.3	3.4
Mn(0) Gal(30)	54.8	44.6	0.12	0.08	6.7	3.8
Mn(0) Gal(60)	50.9	26.6	0.11	0.13	5.2	3.4
Mn(25) Gal(15)	53.1	54.1	0.11	0.06	5.7	3.0
Mn(50) Gal(30)	54.3	51.2	0.11	0.06	5.5	3.1

^aMn concentrations are given in μ M; Gal concentrations are given in mM.

(Figure 4A). The non-glycosylated protein was <1% in control and most supplemented cultures, ~2% with 50 μ M Mn, 1% with 60 mM Gal or 50 μ M Mn + 30 mM Gal, and ~4% in originator Humira.

In clone 14, Mn and/or Gal supplementation similarly altered glycan profiles (Figure 4B, Table 3), reducing G0F. A synergistic effect was observed with 50 μ M Mn + 30 mM Gal, decreasing G0F by 27%, 7% above the sum of individual effects (5% Mn, 15% Gal). An additive effect was seen with 25 μ M Mn + 15 mM Gal, reducing G0F by 23%, matching the sum of individual decreases. In clone 14, G1F increased synergistically with 50 μ M

Mn + 30 mM Gal and additively with 25 μ M Mn + 15 mM Gal. Supplementation with 0–60 mM Gal and/or 0–75 μ M Mn increased G1F, G2F, and Man6, while reducing -NAF. Man5 increased in most conditions, especially with combined Mn + Gal (25 μ M + 15 mM or 50 μ M + 30 mM). 60 mM Gal alone elevated G1F, Man6-8, and reduced G0F more than other conditions. Non-glycosylated protein remained <1% across all supplemented cultures.

To assess transcriptional effects of 50 μ M Mn, HC and LC mRNA levels in clone 9 were measured via qRT-PCR (Figure S4). No significant differences were observed compared to control.

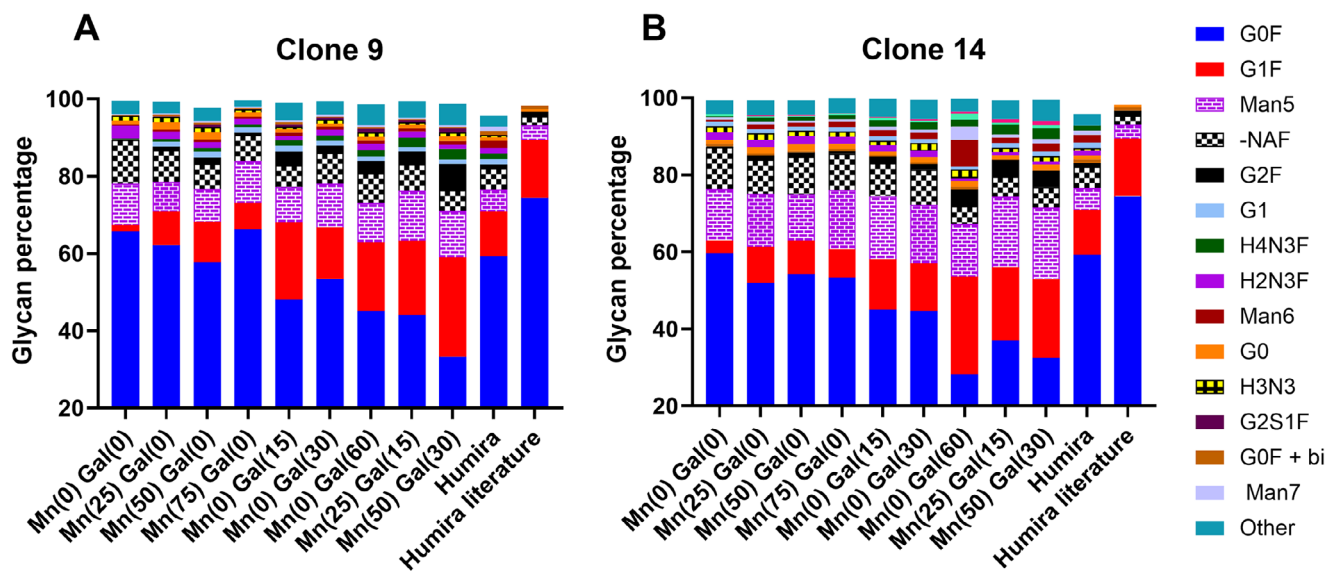


FIGURE 4 | Glycosylation profiles determined by glycopeptide mass spectrometry of clone 9 (A) and clone 14 (B) in cultures supplemented with manganese and/or galactose compared with glycan profiles of originator Humira and reported in the literature [17].

TABLE 3 | Variation in the most significant glycans in adalimumab produced by clones 9 and 14 in cultures supplemented with manganese and/or galactose.

Glycan	G0F			G1F		Man5		-NAF		G2F	
Culture condition	%	% change	%	% change	%	% change	%	% change	%	% change	
		from control		from control		from control		from control		from control	
Clone 9											
Mn(0 ^a) Gal(0)	66	N/A	2	N/A	11	N/A	11	N/A	0.1	N/A	
Mn(25) Gal(0)	62	−3.5	9	7	8	−3	8	−3.2	1	0.9	
Mn(50) Gal(0)	58	−8	10	9	9	−2	7	−5	2	1.5	
Mn(75) Gal(0)	66	0.4	7	5.2	11	0.3	7	−5	0.6	0.5	
Mn(0) Gal(15)	48	−18	20	18	9	−1.5	6	−6	4	3.5	
Mn(0) Gal(30)	54	−12.2	13	12	11	0.8	8	−3.5	2	2	
Mn(0) Gal(60)	45	−21	18	16.2	10	−0.4	8	−4	3	3	
Mn(25) Gal(15)	44	−22	19	17.5	13	2	7	−4.5	3	3.2	
Mn(50) Gal(30)	33	−32.5	26	24	12	1	5	−6	7	7	
Clone 14											
Mn(0) Gal(0)	60	N/A	3	N/A	13	N/A	11	N/A	0.3	N/A	
Mn(25) Gal(0)	52	−8	9	6	14	0.4	9	−2	1	0.7	
Mn(50) Gal(0)	54	−5	9	5	12	−1	10	−0.9	1	0.8	
Mn(75) Gal(0)	53	−6.2	7	4	15	2	9	−1	0.7	0.4	
Mn(0) Gal(15)	45	−14.5	13	10	16	3	9	−2	2	1	
Mn(0) Gal(30)	45	−15	12	9	15	2	9	−1.5	2	1.2	
Mn(0) Gal(60)	28	−31	25	22	14	0.4	5	−6.2	4	4	
Mn(25) Gal(15)	37	−22.5	19	16	18	5	5	−5.5	4	4	
Mn(50) Gal(30)	33	−27	21	17	19	5.2	5	−5.5	4	4	

^aMn concentrations are given in μ M; Gal concentrations are given in mM.

Control cultures showed increased mRNA levels from Days 6 to 9, peaking at Day 9. Mn-treated cells exhibited a slight, non-significant decrease at Day 9. Reductions in titer and Qp likely reflect this minor change in mRNA. qRT-PCR analysis of selected N-glycosylation genes on Days 6 and 9 (Figure S5) showed that 50 μ M Mn increased Mgat3 and Mgat5b on Day 6 compared with control cultures, though threshold cycles were high ($\sim$ 30), indicating low expression. No significant changes were observed for Mgat2, Uap1, Gmds, DPAGT1, Slc35a2, Slc35c1, ST3Gal4, or ST3Gal6 (Figures S5A and S5B). Expression profiles varied over time in both control and Mn-treated cultures.

No correlation was found between glycosylation-related gene expression and glycan changes under 50 μ M Mn. G1F and G2F increased, yet B4GalT1-2 and Slc35a2 showed no differential expression; B4GalT3 decreased on Day 9. G0F decreased without changes in Mgat1, Mgat2, Uap1, or Slc35a3. Man5 slightly decreased, though Man1a1, Man1a2, Man1c1, Man2a1, and Man2a2 were unchanged. Hybrid glycans (-NAF) declined, despite no differential expression in genes responsible for -NAF formation (Mgat1, Mgat2, Man2a1, Man2a2, FUT8, Slc35c1).

4 | Model-Based Analysis of the Effects of Mn and Gal Feeding

We had hypothesized that increasing the galactosylation would shift the equilibrium towards more mature glycans, thereby reducing high mannose and hybrid species. However, despite increased galactosylation, reductions in high mannose glycans were modest, and in some cases, levels increased.

4.1 | Mathematical Model Definition

To elucidate the mechanisms underlying increased high mannose species and to develop a predictive approach for controlling mAb glycosylation, we applied the dynamic mathematical model of glycosylation described by Jimenez del Val et al. [46]. The model incorporates 77 glycan structures undergoing 96 reactions catalyzed by seven enzymes: ManI, GnTI (MgatI), ManII, GnTII (MgatII), Fut8, B4GalT, and ST3Gal. The model assumes Golgi cisternal maturation, in which the Golgi compartments are synthesized from components of the endoplasmic reticulum and transit towards the secretory compartments of the cell [62, 63]. Golgi-resident proteins recycle via retrograde transport mediated by COPI-coated vesicles [64, 65]. Accordingly, the Golgi apparatus was modeled as a plug flow reactor. Assumptions included no axial distribution, constant linear velocity, no mass transfer limitations, and constant Golgi compartment diameter. Model equations, including material balances and enzyme kinetics, are provided in Equations (1)–(7) and Table S3 (Modeling Supporting Information). Parameters were estimated to define enzyme concentration profiles along the Golgi and enzyme dissociation constants (K_d). Calculated enzyme concentration profiles are shown in Figure S7, and dissociation constants are listed in Table S4. The predicted glycoform distribution closely matched the experimentally observed distribution for clone 9 control cultures, demonstrating the model's utility (Figure S8). Simulations were performed using gPROMS Process Release 2023.2.1 [66] running on a desktop PC with a Core i7 processor and 16 GB RAM.

4.1.1 | Mn Feeding

Manganese (Mn) is a known cofactor of GnTI (MgatI), GnTII (MgatII), and B4GalT enzymes. We hypothesized that Mn supplementation would increase the turnover rates (rate of the catalytic step) of these enzymes, altering the glycoform distribution [67]. Based on this hypothesis, the turnover rates for the GnTI, GnTII, and B4GalT enzymes were estimated for Mn-fed samples by attempting to match the model output with the experimentally observed glycoform distribution.

4.1.2 | Gal Feeding

To evaluate the effect of galactose (Gal) feeding during culture, we hypothesized that Gal addition increases intracellular UDP-Gal, the co-substrate required for mAb galactosylation reactions. With this hypothesis, parameter estimation was used to calculate UDP-Gal concentrations in Gal-fed samples. Based on the nucleotide sugar donor (NSD) biosynthetic pathway (Figure 5E), we assumed that the sum of the UDP-GlcNAc and UDP-Gal concentrations was constant, based on the UTP requirement for synthesis of both NSD co-substrates. From a modeling perspective, this assumption implies that an increase in one NSD necessitates a decrease in the other. For the Gal feeding simulations, enzyme concentrations, turnover rates, and dissociation constants were fixed at the control culture values.

4.2 | Modeling Results

4.2.1 | Mn Feeding

Figure 5A compares the calculated and experimental glycoform distributions, while Figure 5B presents the estimated turnover rates for GnTI, GnTII, and B4GalT. The model predictions agreed well with experimental data, with the largest deviations observed in G1F and Man5 species. Mn addition increased the turnover rate of B4GalT, consistent with the initial hypothesis, except at 75 μ M Mn, where fewer galactosylated species were observed (Figure 5B). In contrast, the turnover rates (k_f) for GnTI and GnTII remained at nominal (literature) values, indicating no catalytic enhancement by Mn. These results were unexpected as the optimal Mn concentrations for GnTI and GnTII are between 10 and 80 mM [68, 69], whereas the Mn concentration in the cultures is three orders of magnitude lower (Figure S6).

25- and 50- μ M Mn feeds decreased Man5 species for clone 9 by 2% and 3%, respectively (Table 3). These cultures showed Qp values of 0.11 picograms/cell/day (pcd) and 0.13 pcd, which are 35% and 23% lower than the control culture (0.17 pcd). Under cisternal maturation, reduced Qp increases Golgi residence time, leading to increased glycan processing for the same enzymatic activity (as represented by k_f). Therefore, the estimated k_f values for GnTI and GnTII remain at the lower bounds (nominal literature values) because reduced Qp, not increased enzymatic activity reduces the Man5 glycans. This result is consistent with previous studies linking increased Man5 secretion with hypothermia-induced increases in Qp [45, 70].

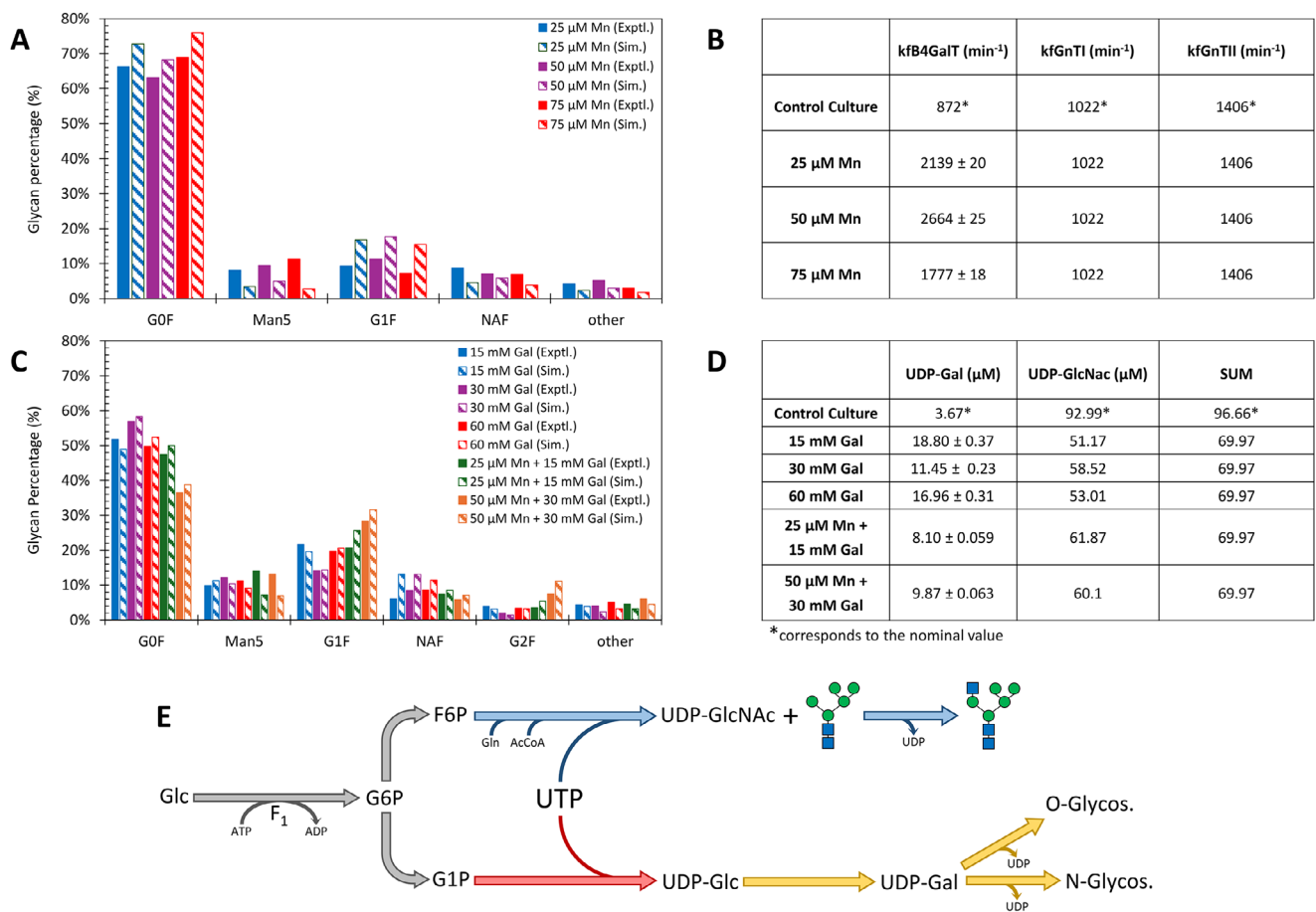


FIGURE 5 | Modeling results from analysis of clone 9 cultures supplemented with manganese and/or galactose. (A) Comparison of calculated vs. experimental glycan distribution for samples with manganese addition. (B) Estimated values of turnover rates for samples with manganese addition. (C) Comparison of calculated vs. experimental glycan distribution for samples with galactose addition. (D) Estimated values for UDP-Gal and UDP-GlcNAc for samples with galactose addition. (E) Biosynthesis of UDP-GlcNAc, UDP-Glc, and UDP-Gal. The synthesis of UDP-GlcNAc and UDP-Gal relies on a common intracellular pool of uridine triphosphate (UTP). If UTP is limiting and more UDP-Gal is consumed towards protein glycosylation, then less UDP-GlcNAc would be available for the consumption of Man5 towards more processed glycans.

4.2.2 | Gal Feeding

The estimated values for UDP-Gal concentration are shown in Figure 5D, and Figure 5C compares the calculated and experimental glycoform distributions. The model closely reproduces the experimental data using the estimated UDP-Gal and UDP-GlcNAc concentrations, with larger discrepancies occurring in G0F and Man5 species. Our results indicate Gal supplementation increases UDP-Gal concentrations, reducing UDP-GlcNAc availability, promoting Man5 production.

5 | Discussion

In this study, two cell clones, clone 9 and clone 14, derived in the same manner, from the same parental CHO cell line and producing the same recombinant monoclonal antibody, exhibited different growth, productivity, and glycosylation profiles under identical culture conditions. Both clones exhibited undesirable glycans, particularly excessively high mannose (Man5) and hybrid (e.g., -NAF) glycans, which can be immunogenic and have higher clearance rates.

The glycosylation patterns of clones 9 and 14 differed from the originator Humira in our measurements and from literature-reported Humira glycan profiles. Variations in hosts, cell line selection, process parameters, and media composition can cause significant differences in mAb glycosylation between originator products and biosimilars [16, 31, 71]. mAb glycoform variations affect biological activity, immunogenicity, efficacy, and safety. For example, removal of the Fc glycan core fucose residue significantly alters antibody-dependent cell-mediated cytotoxicity [72–74]. Changes in terminal Gal affect complement-dependent cytotoxicity (CDC) [75, 76], and terminal mannose or sialic acid modulate the glycoprotein circulatory half-life [77]. Thus, controlling process conditions and cell culture media components is critical to maintaining a consistent glycoform profile.

We examined the impact of Mn and Gal supplementation on glycan profiles. Cultures supplemented with Mn and/or Gal exhibited similar changes across clones 9 (Figure 4A, Table 3) and 14 (Figure 4B, Table 3), with increases in G1F and G2F species and reductions in G0F and -NAF species. For clone 9, 25 μM and 50 μM Mn reduced Man5 species and increased G1F and G2F species. Clone 14 showed similar results with 50 μM Mn supplementation. While we saw a significant increase in galac-

tosylation at Mn concentrations up to 50 μM , other investigators saw improvement at significantly lower concentrations (up to 1 μM) with greater decreases in high mannose glycans [38]. Mn concentrations up to 1 μM also decreased high mannose glycans in studies by Pacis et al., although they did not examine effects on galactosylation [39]. The impact of Mn on galactosylation is likely due to its role as a metal ion cofactor that enhances B4GalT1 galactosylation activity [13, 78, 79]. Gal supplementation led to comparable glycosylation changes in both clones, with marked gains in G1F and G2F species at the expense of G0F species. Presumably, adding Gal directly affects UDP-Gal availability for galactosyltransferases in the Golgi [80, 81]. However, the effects of Gal on G0F, G1F, and G2F were not proportional to Gal concentration or monotonic. Supplementation with both Mn and Gal increased galactosylation synergistically, but paradoxically, high mannose glycans increased under some conditions.

Glycosylation gene expression showed few differences between clone 9 and clone 14 or upon supplementation with 50 μM Mn. No correlation was observed between differential gene expression and glycosylation patterns, suggesting that enzyme activity and substrate availability likely drive the observed changes. Both clones expressed primarily core-fucosylated, complex, biantennary Fc glycans, mainly G0F, with clone 9 showing higher G0F levels than clone 14. *Mgat1* and *Mgat2*, the genes encoding for enzymes responsible for synthesizing bi-antennary N-linked glycans (GnTI and GnTII), [81, 82] exhibited no differential expression between the clones. GnTI reduces high mannose glycans [39], and previous studies have inactivated GnTI (*Mgat1*) to produce high mannose glycans [83]. Here, clones 9 and 14 exhibited comparable *Mgat1* expression levels compared to the other glycosylation enzymes based on threshold cycle, but excessive Man5 glycans were still produced. High mannose levels can be reduced by increasing *Mgat1* and mannosidase activity [13]. *Mgat1*, *Man1a1*, *Man1a2*, *Man1c1*, *Man2a1*, and *Man2a2* expression levels were similar for both control and 50 μM Mn conditions on Days 6 and 9, suggesting that adding Mn did not affect their transcription. Nonetheless, 50 μM Mn slightly decreased Man5 and increased G1F and G2F without changing B4GalT1-2, indicating Mn altered glycan profiles without changing transcript levels.

The Golgi glycosylation model identified mechanisms underlying glycoform distribution shifts upon Mn and Gal feeding. Specifically, increased turnover rate (k_f) is the mechanism whereby Mn feeding increases mAb galactosylation. Moreover, increased k_f and UDP-Gal availability combine to increase galactosylation when Mn and Gal are simultaneously fed. These results are consistent with previous work [47], in which a computational model was used to define an optimal uridine/galactose feeding strategy, which, along with 1 μM Mn supplementation, maximized mAb galactosylation while minimizing negative impacts on cell growth and productivity. In contrast, Mn supplementation slightly decreased Man5 production, caused by increased residence time in Golgi due to lower Q_p , not increased GnT1 k_f . This observation is consistent with previous work showing that increased Q_p caused by mild hypothermic conditions drove secretion of Man5 mAb glycoforms [45, 70]. Together with the inverse relationship between galactosylation and Man5 production observed in clone 14 cultures, these results suggest that increased galactosylation increases demand for UDP-Gal

depleting UTP availability for UDP-GlcNAc biosynthesis, thereby increasing Man5 production (Figure 5E). Previous work [47], extended the Golgi glycosylation model to capture (i) cell culture dynamics, (ii) intracellular nucleotide sugar metabolism, and (iii) the effects of uridine, manganese, and galactose (UMG) feeding on both culture dynamics and glycosylation. With further process characterization—such as experiments tracking cell culture dynamics, intracellular nucleotide sugar pools, and mAb glycosylation profiles under different UMG regimens—the model can be parameterized. Once parameterized, dynamic optimization can be applied to determine the timing and dose of UMG feeds that achieve a desired glycosylation profile while minimizing impacts on other KPIs, such as cell growth and productivity.

6 | Conclusions

This study shows how Mn and Gal modulate mAb glycosylation patterns in specific CHO clones during fed-batch experiments. These findings also point to supplementation strategies to adjust glycan distributions toward desired glycoforms. Feeding Gal and Mn increased mAb galactosylation, presumably due to enhanced UDP-Gal biosynthesis. Computational results suggest that increased UDP-Gal synthesis may deplete UTP availability for UDP-GlcNAc production, leading to Man5 glycan accumulation. This UTP-Man5 link is consistent with previous uridine-Mn-Gal feeding strategies [79, 84], which included uridine to prevent UTP depletion. Further work is needed to achieve consistent, high-quality N-glycosylation profiles in therapeutic mAbs. Both glycoengineering and process engineering offer promising paths forward.

Author Contributions

R.P. performed the experiments, devised experimental methods, and drafted the manuscript. S.G. performed the modeling studies and drafted the modeling methods and results. M.Y. provided project conception and financial support and reviewed the manuscript. I.J.V. conceived the modeling studies, provided financial support, and revised the manuscript. S.T.S. designed the project, provided experimental guidance, provided financial support, and revised the manuscript. During revision, S.T.S. applied ChatGPT 5.0 to improve clarity and conciseness.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Mass spectrometry data are included as supplementary data. All other data is available upon request.

References

1. A. G. Vulto and O. A. Jaquez, "The Process Defines the Product: What Really Matters in Biosimilar Design and Production?," *Rheumatology* 56, no. 4 (2017): iv14–iv29, <https://doi.org/10.1093/rheumatology/kex278>.
2. L. Hajba, Á. Szekrényes, B. Borza, and A. Guttman, "On the Glycosylation Aspects of Biosimilarity," *Drug Discovery Today* 23, no. 3 (2018): 616–625, <https://doi.org/10.1016/j.drudis.2018.01.009>.
3. B. S. Sekhon and V. Saluja, "Biosimilars: An Overview," *Biosimilars* 1, (2011): 1–11, <https://doi.org/10.2147/BS.S16120>.
4. J. M. Reichert, "Monoclonal Antibodies as Innovative Therapeutics," *Current Pharmaceutical Biotechnology* 9, no. 6 (2008): 423–430, <https://doi.org/10.2174/138920108786786358>.
5. P. Chames, M. Van Regenmortel, E. Weiss, and D. Baty, "Therapeutic Antibodies: Successes, Limitations and Hopes for the Future," *British Journal of Pharmacology* 157, no. 2 (2009): 220–233, <https://doi.org/10.1111/j.1476-5381.2009.00190.x>.
6. F. Cymer, H. Beck, A. Rohde, and D. Reusch, "Therapeutic Monoclonal Antibody N-Glycosylation—Structure, Function and Therapeutic Potential," *Biologicals* 52 (2018): 1–11, <https://doi.org/10.1016/j.biologics.2017.11.001>.
7. L. K. Hmiel, K. A. Brorson, and M. T. Boyne, "Post-Translational Structural Modifications of Immunoglobulin g and Their Effect on Biological Activity," *Analytical and Bioanalytical Chemistry* 407, no. 1 (2015): 79–94, <https://doi.org/10.1007/s00216-014-8108-x>.
8. A. A. Shukla, B. Hubbard, T. Tressel, S. Guhan, and D. Low, "Downstream Processing of Monoclonal Antibodies—Application of Platform Approaches," *Journal of Chromatography B, Analytical Technologies in the Biomedical and Life Sciences* 848, no. 1 (2007): 28–39, <https://doi.org/10.1016/j.jchromb.2006.09.026>.
9. Y. Durocher and M. Butler, "Expression Systems for Therapeutic Glycoprotein Production," *Current Opinion in Biotechnology* 20, no. 6 (2009): 700–707, <https://doi.org/10.1016/j.copbio.2009.10.008>.
10. H. Dahodwala, M. Nowey, T. Mitina, and S. T. Sharfstein, "Effects of Clonal Variation on Growth, Metabolism, and Productivity in Response to Trophic Factor Stimulation: A Study of Chinese Hamster Ovary Cells Producing a Recombinant Monoclonal Antibody," *Cytotechnology* 64, no. 1 (2012): 27–41, <https://doi.org/10.1007/s10616-011-9388-z>.
11. J. Y. Kim, Y. G. Kim, and G. M. Lee, "CHO Cells in Biotechnology for Production of Recombinant Proteins: Current State and Further Potential," *Applied Microbiology and Biotechnology* 93, no. 3 (2012): 917–930, <https://doi.org/10.1007/s00253-011-3758-5>.
12. V. Tejwani, M. R. Andersen, J. H. Nam, and S. T. Sharfstein, "Glycoengineering in CHO Cells: Advances in Systems Biology," *Biotechnology Journal* 13, no. 3 (2018): 1700234, <https://doi.org/10.1002/biot.201700234>.
13. H. H. Shi and C. T. Goudar, "Recent Advances in the Understanding of Biological Implications and Modulation Methodologies of Monoclonal Antibody N-Linked High Mannose Glycans," *Biotechnology and Bioengineering* 111, no. 10 (2014): 1907–1919, <https://doi.org/10.1002/bit.25318>.
14. P. M. Rudd, T. Elliott, P. Cresswell, I. A. Wilson, and R. A. Dwek, "Glycosylation and the Immune System," *Science* 291, no. 5512 (2001): 2370–2376, <https://doi.org/10.1126/science.291.5512.2370>.
15. P. Zhang, S. Woen, T. Wang, et al., "Challenges of Glycosylation Analysis and Control: An Integrated Approach to Producing Optimal and Consistent Therapeutic Drugs," *Drug Discovery Today* 21, no. 5 (2016): 740–765, <https://doi.org/10.1016/j.drudis.2016.01.006>.
16. H. Dahodwala and S. T. Sharfstein, "Biosimilars: Imitation Games," *ACS Medicinal Chemistry Letters* 8, no. 7 (2017): 690–693, <https://doi.org/10.1021/acsmmedchemlett.7b00199>.
17. S. Bandyopadhyay, M. Mahajan, T. Mehta, et al., "Physicochemical and Functional Characterization of a Biosimilar Adalimumab ZRC-3197," *Biosimilars* 5, no. 1 (2014): 1–18, <https://doi.org/10.2147/bs.S75573>.
18. M. Weise, M. C. Bielsky, K. De Smet, et al., "Biosimilars—Why Terminology Matters," *Nature Biotechnology* 29, no. 8 (2011): 690–693, <https://doi.org/10.1038/nbt.1936>.
19. M. S. Dordal, F. F. Wang, and E. Goldwasser, "The Role of Carbohydrate in Erythropoietin Action*," *Endocrinology* 116, no. 6 (1985): 2293–2299, <https://doi.org/10.1210/endo-116-6-2293>.
20. M. Nose and H. Wigzell, "Biological Significance of Carbohydrate Chains on Monoclonal Antibodies," *Proceedings of the National Academy of Sciences of the United States of America* 80, no. 21 (1983): 6632–6636, <https://doi.org/10.1073/pnas.80.21.6632>.
21. V. Azevedo, B. Hassett, J. E. Fonseca, et al., "Differentiating Biosimilarity and Comparability in Biotherapeutics," *Clinical Rheumatology* 35, no. 12 (2016): 2877–2886, <https://doi.org/10.1007/s10067-016-3427-2>.
22. R. Pranomphon, V. Tejwani, H. Dahodwala, M. Yamabhai, and S. T. Sharfstein, "Metabolic and Process Engineering to Control Glycan Structures for Biopharmaceuticals Produced in Cultured Mammalian Cells," in *Bioprocessing, Bioengineering and Process Chemistry in the Biopharmaceutical Industry: Using Chemistry and Bioengineering to Improve the Performance of Biologics*, ed. K. Gadamasetti and S. A. Kolodziej (Springer Nature, 2024), 135–168, https://doi.org/10.1007/978-3-031-62007-2_6.
23. U.S. Food and Drug Administration. (2025). Retrieved 05/29/2025 from, <https://www.fda.gov/drugs/biosimilars/biosimilar-product-information>.
24. G. Xu and J. Wang, "The Current Status of the Biosimilars Landscape in China," *Biologicals* 85 (2024): 101744, <https://doi.org/10.1016/j.biologics.2024.101744>.
25. D. Gao, L. Nie, J. Yuan, et al., "Physicochemical and Functional Characterization of HS016, a Biosimilar of Adalimumab (Humira)," *Journal of Pharmaceutical Sciences* 111, no. 4 (2022): 1142–1151, <https://doi.org/10.1016/j.xphs.2021.12.001>.
26. J. C. Kwon, O. H. Kwon, R. U. Jeong, et al., "Physicochemical and Biological Similarity Assessment of LBAL, a Biosimilar to Adalimumab Reference Product (Humira(R))." *Anim Cells Syst (Seoul)* 25, no. 3 (2021): 182–194, <https://doi.org/10.1080/19768354.2021.1943709>.
27. N. Lee, J. J. Lee, H. Yang, et al., "Evaluation of Similar Quality Attribute Characteristics in SB5 and Reference Product of Adalimumab," *MAbs* 11, no. 1 (2019): 129–144, <https://doi.org/10.1080/19420862.2018.1530920>.
28. J. Liu, T. Eris, C. Li, S. Cao, and S. Kuhns, "Assessing Analytical Similarity of Proposed Amgen Biosimilar ABP 501 to Adalimumab," *BioDrugs* 30, no. 4 (2016): 321–338, <https://doi.org/10.1007/s40259-016-0184-3>.
29. L. Magnenat, A. Palmese, C. Fremaux, et al., "Demonstration of Physicochemical and Functional Similarity Between the Proposed Biosimilar Adalimumab MSB11022 and Humira®," *MAbs* 9, no. 1 (2017): 127–139, <https://doi.org/10.1080/19420862.2016.1259046>.
30. E. Zhang, L. Xie, P. Qin, et al., "Quality by Design–Based Assessment for Analytical Similarity of Adalimumab Biosimilar HLX03 to Humira®," *AAPS Journal [Electronic Resource]* 22, no. 3 (2020): 69, <https://doi.org/10.1208/s12248-020-00454-z>.
31. E. Edwards, M. Livanos, A. Krueger, et al., "Strategies to Control Therapeutic Antibody Glycosylation During Bioprocessing: Synthesis and Separation," *Biotechnology and Bioengineering* 119, no. 6 (2022): 1343–1358, <https://doi.org/10.1002/bit.28066>.
32. J. Xu, Z. Shao, Z. Wang, Y. Huang, X. Zou, and Y. Shen, "Developing a medium combination to attain similar glycosylation profile to originator by DoE and cluster analysis method," *Scientific Reports*, 11, no. 1 (2021), <https://doi.org/10.1038/s41598-021-86447-0>.

33. S. Rameez, Y. K. Gowtham, G. Nayar, and S. S. Mostafa, "Modulation of High Mannose Levels in N-Linked Glycosylation Through Cell Culture Process Conditions to Increase Antibody-Dependent Cell-Mediated Cytotoxicity Activity for an Antibody Biosimilar," *Biotechnology Progress* 37, no. 5 (2021): 3176, <https://doi.org/10.1002/btpr.3176>.
34. N. Gangwar, N. Dixit, and A. S. Rathore, "N-Glycosylation Modulators for Targeted Manipulation of Glycosylation for Monoclonal Antibodies," *Applied Microbiology and Biotechnology* 109, no. 1 (2025): 16, <https://doi.org/10.1007/s00253-025-13405-5>.
35. J. Kranjc, L. Kramer, M. Mikelj, M. Anderluh, A. Pisljar, and M. Brinc, "Modulating Antibody N-Glycosylation Through Feed Additives Using a Multi-Tiered Approach," *Frontiers in Bioengineering and Biotechnology* 12 (2024): 1448925, <https://doi.org/10.3389/fbioe.2024.1448925>.
36. J. Ehret, M. Zimmermann, T. Eichhorn, and A. Zimmer, "Impact of Cell Culture Media Additives on IgG Glycosylation Produced in Chinese Hamster Ovary Cells," *Biotechnology and Bioengineering* 116, no. 4 (2019): 816–830, <https://doi.org/10.1002/bit.26904>.
37. C. Benga, L. M. Rives, and P. Hossler (2016). *Methods for Controlling the Galactosylation Profile of Recombinantly Expressed Proteins* (United States Patent No. US 9,365,645 B1).
38. S. Loeblich, E. Clark, K. Ladd, et al., "Comprehensive Manipulation of Glycosylation Profiles Across Development Scales," *MAbs* 11, no. 2 (2019): 335–349, <https://doi.org/10.1080/19420862.2018.1527665>.
39. E. Pacis, M. Yu, J. Autsen, R. Bayer, and F. Li, "Effects of Cell Culture Conditions on Antibody N-Linked Glycosylation—What Affects High Mannose 5 Glycoform," *Biotechnology and Bioengineering* 108, no. 10 (2011): 2348–2358, <https://doi.org/10.1002/bit.23200>.
40. F. J. Krambeck, S. V. Bennun, M. R. Andersen, and M. J. Betenbaugh, "Model-Based Analysis of N-Glycosylation in Chinese Hamster Ovary Cells," *PLoS ONE* 12, no. 5 (2017): 0175376, <https://doi.org/10.1371/journal.pone.0175376>.
41. F. J. Krambeck and M. J. Betenbaugh, "A Mathematical Model of N-Linked Glycosylation," *Biotechnology and Bioengineering* 92, no. 6 (2005): 711–728, <https://doi.org/10.1002/bit.20645>.
42. J. V. Reddy, T. Leibiger, S. K. Singh, K. H. Lee, E. Papoutsakis, and M. Ierapetritou, "A Novel, Site-Specific N-Linked Glycosylation Model Provides Mechanistic Insights Into the Process-Condition Dependent Distinct Fab and Fc Glycosylation of an IgG1 Monoclonal Antibody Produced by CHO VRC01 Cells," *Biotechnology and Bioengineering* 122, no. 4 (2025): 761–778, <https://doi.org/10.1002/bit.28916>.
43. P. Umaña and J. E. Bailey, "A Mathematical Model of N-Linked Glycoform Biosynthesis [Article]," *Biotechnology and Bioengineering* 55, no. 6 (1997): 890–908, [https://doi.org/10.1002/\(SICI\)1097-0290\(19970920\)55:6<890::AID-BIT7>3.0.CO;2-B](https://doi.org/10.1002/(SICI)1097-0290(19970920)55:6<890::AID-BIT7>3.0.CO;2-B).
44. P. Hossler, B. C. Mulukutla, and W. S. Hu, "Systems Analysis of N-Glycan Processing in Mammalian Cells," *PLoS ONE* 2, no. 8 (2007): 713, <https://doi.org/10.1371/journal.pone.0000713>.
45. I. Jimenez del Val, Y. Fan, and D. Weilguny, "Dynamics of Immature mAb Glycoform Secretion during CHO Cell Culture: An Integrated Modelling Framework," *Biotechnology Journal* 11, no. 5 (2016): 610–623, <https://doi.org/10.1002/biot.201400663>.
46. I. Jimenez del Val, J. M. Nagy, and C. Kontoravdi, "A Dynamic Mathematical Model for Monoclonal Antibody N-Linked Glycosylation and Nucleotide Sugar Donor Transport Within a Maturing Golgi Apparatus," *Biotechnology Progress* 27, no. 6 (2011): 1730–1743, <https://doi.org/10.1002/btpr.688>.
47. P. Kotidis, P. Jedrzejewski, S. N. Sou, et al., "Model-Based Optimization of Antibody Galactosylation in CHO Cell Culture," *Biotechnology and Bioengineering* 116, no. 7 (2019): 1612–1626, <https://doi.org/10.1002/bit.26960>.
48. S. Hutter, T. K. Villiger, D. Brühlmann, et al., "Glycosylation Flux Analysis Reveals Dynamic Changes of Intracellular Glycosylation Flux Distribution in Chinese Hamster Ovary Fed-Batch Cultures," *Metabolic Engineering* 43, no. Pt A (2017): 9–20, <https://doi.org/10.1016/j.ymben.2017.07.005>.
49. C. Liang, A. W. T. Chiang, A. H. Hansen, et al., "A Markov Model of Glycosylation Elucidates Isozyme Specificity and Glycosyltransferase Interactions for Glycoengineering," *Current Research in Biotechnology* 2 (2020): 22–36, <https://doi.org/10.1016/j.crbiot.2020.01.001>.
50. C. Liang, A. W. T. Chiang, and N. E. Lewis, "GlycoMME, a Markov Modeling Platform for Studying N-Glycosylation Biosynthesis From Glycomics Data," *STAR Protocols* 4, no. 2 (2023): 102244, <https://doi.org/10.1016/j.xpro.2023.102244>.
51. A. Antonakoudis, B. Strain, R. Barbosa, I. Jimenez del Val, and C. Kontoravdi, "Synergising Stoichiometric Modelling With Artificial Neural Networks to Predict Antibody Glycosylation Patterns in Chinese Hamster Ovary Cells," *Computers & Chemical Engineering* 154 (2021): 107471, <https://doi.org/10.1016/j.compchemeng.2021.107471>.
52. P. Kotidis and C. Kontoravdi, "Harnessing the Potential of Artificial Neural Networks for Predicting Protein Glycosylation," *Metabolic Engineering Communications* 10 (2020): 00131, <https://doi.org/10.1016/j.mec.2020.e00131>.
53. P. Seber and R. D. Braatz, "Linear and Neural Network Models for Predicting N-Glycosylation in Chinese Hamster Ovary Cells Based on B4GALT Levels," *Computers & Chemical Engineering* 194 (2025): 108937, <https://doi.org/10.1016/j.compchemeng.2024.108937>.
54. O. Yang and M. Ierapetritou, "Mab Production Modeling and Design Space Evaluation Including Glycosylation Process," *Processes* 9, no. 2 (2021): 324, <https://www.mdpi.com/2227-9717/9/2/324>.
55. J. V. Reddy, K. Raudenbush, E. T. Papoutsakis, and M. Ierapetritou, "Cell-Culture Process Optimization via Model-Based Predictions of Metabolism and Protein Glycosylation," *Biotechnology Advances* 67 (2023): 108179, <https://doi.org/10.1016/j.biotechadv.2023.108179>.
56. V. Muralidharan-Chari, Z. Wurz, F. Doyle, et al., "PTSelect™: A Post-Transcriptional Technology That Enables Rapid Establishment of Stable CHO Cell Lines and Surveillance of Clonal Variation," *Journal of Biotechnology* 325 (2021): 360–371, <https://doi.org/10.1016/j.jbiotec.2020.09.025>.
57. J. Y. Baik, H. Dahodwala, E. Oduah, et al., "Optimization of Bioprocess Conditions Improves Production of a CHO Cell-Derived, Bioengineered Heparin," *Biotechnology Journal* 10, no. 7 (2015): 1067–1081, <https://doi.org/10.1002/biot.201400665>.
58. K. J. Livak and T. D. Schmittgen, "Analysis of Relative Gene Expression Data Using Real-Time Quantitative PCR and the $2^{-\Delta\Delta CT}$ Method," *Methods (San Diego, Calif)* 25, no. 4 (2001): 402–408, <https://doi.org/10.1006/meth.2001.1262>.
59. J. Peng, S. M. Patil, D. A. Keire, and K. Chen, "Chemical Structure and Composition of Major Glycans Covalently Linked to Therapeutic Monoclonal Antibodies by Middle-Down Nuclear Magnetic Resonance," *Analytical Chemistry* 90, no. 18 (2018): 11016–11024, <https://doi.org/10.1021/acs.analchem.8b02637>.
60. A. Planinc, B. Dejaegher, Y. Vander Heyden, et al., "Batch-to-Batch N-Glycosylation Study of Infliximab, Trastuzumab and Bevacizumab, and Stability Study of Bevacizumab," *European Journal of Hospital Pharmacy* 24, no. 5 (2017): 286–292, <https://doi.org/10.1136/ejhpharm-2016-001022>.
61. S. Sha, C. Agarabi, K. Brorson, D.-Y. Lee, and S. Yoon, "N-Glycosylation Design and Control of Therapeutic Monoclonal Antibodies," *Trends in Biotechnology* 34, no. 10 (2016): 835–846, <https://doi.org/10.1016/j.tibtech.2016.02.013>.
62. B. S. Glick and V. Malhotra, "The Curious Status of the Golgi Apparatus," *Cell* 95, no. 7 (1998): 883–889, [https://doi.org/10.1016/S0092-8674\(00\)81713-4](https://doi.org/10.1016/S0092-8674(00)81713-4).
63. B. S. Glick and A. Nakano, "Membrane Traffic Within the Golgi Apparatus," *Annual Review of Cell and Developmental Biology* 25 (2009): 113–132, <https://doi.org/10.1146/annurev.cellbio.24.110707.175421>.

64. T. Nilsson, C. E. Au, and J. J. Bergeron, "Sorting out Glycosylation Enzymes in the Golgi Apparatus," *FEBS Letters* 583, no. 23 (2009): 3764–3769, <https://doi.org/10.1016/j.febslet.2009.10.064>.
65. H. R. Pelham, "Getting through the Golgi Complex," *Trends in Cell Biology* 8, no. 1 (1998): 45–49, [https://doi.org/10.1016/s0962-8924\(97\)01185-9](https://doi.org/10.1016/s0962-8924(97)01185-9).
66. Siemens Industry Software. (2024). Gproms Process Release 202321, <http://www.siemens.com/global/en/products/automation/industry-software/gproms-digital-process-design-and-operations/gproms-modelling-environments/gproms-process.html>, 1997–2024.
67. D. L. Witsell, C. E. Casey, and M. C. Neville, "Divalent Cation Activation of Galactosyltransferase in Native Mammary Golgi Vesicles," *Journal of Biological Chemistry* 265, no. 26 (1990): 15731–15737, [https://doi.org/10.1016/S0021-9258\(18\)55458-2](https://doi.org/10.1016/S0021-9258(18)55458-2).
68. J. Nishii, N. Kioka, T. Fukada, H. Sakai, and T. Komano, "Characterization of Rat N-Acetylglucosaminyltransferase I Expressed in *Escherichia coli*," *Bioscience, Biotechnology, and Biochemistry* 59, no. 9 (1995): 1750–1752, <https://doi.org/10.1271/bbb.59.1750>.
69. H. Schachter, I. Brockhausen, and E. Hull, "High-Performance Liquid Chromatography Assays for N-Acetylglucosaminyltransferases Involved in N- and O-Glycan Synthesis," *Methods in Enzymology* 179 (1989): 351–397, [https://doi.org/10.1016/0076-6879\(89\)79138-2](https://doi.org/10.1016/0076-6879(89)79138-2).
70. Y. Fan, I. Jimenez Del Val, C. Muller, et al., "A Multi-Pronged Investigation into the Effect of Glucose Starvation and Culture Duration on Fed-Batch CHO Cell Culture," *Biotechnology and Bioengineering* 112, no. 10 (2015): 2172–2184, <https://doi.org/10.1002/bit.25620>.
71. B. L. Duivelshof, W. Jiskoot, A. Beck, J.-L. Veuthey, D. Guilleme, and V. D'Atri, "Glycosylation of Biosimilars: Recent Advances in Analytical Characterization and Clinical Implications," *Analytica Chimica Acta* 1089 (2019): 1–18, <https://doi.org/10.1016/j.aca.2019.08.044>.
72. C. Ferrara, S. Grau, C. Jäger, et al., "Unique Carbohydrate–Carbohydrate Interactions Are Required for High Affinity Binding Between FcγRIII and Antibodies Lacking Core Fucose," *Proceedings of the National Academy of Sciences of the United States of America* 108 (2011): 12669–12674, <https://doi.org/10.1073/pnas.1108455108>.
73. A. Natsume, M. Wakitani, N. Yamane-Ohnuki, et al., "Fucose Removal From Complex-Type Oligosaccharide Enhances the Antibody-Dependent Cellular Cytotoxicity of Single-Gene-Encoded Antibody Comprising a Single-Chain Antibody Linked the Antibody Constant Region," *Journal of Immunological Methods* 306, no. 1 (2005): 93–103, <https://doi.org/10.1016/j.jim.2005.07.025>.
74. T. Shinkawa, K. Nakamura, N. Yamane, et al., "The Absence of Fucose but Not the Presence of Galactose or Bisecting N-Acetylglucosamine of Human IgG1 Complex-Type Oligosaccharides Shows the Critical Role of Enhancing Antibody-Dependent Cellular Cytotoxicity," *Journal of Biological Chemistry* 278 (2003): 3466–3473, <https://doi.org/10.1074/jbc.M210665200>.
75. J. Hodoniczky, Y. Zheng, and D. James, "Control of Recombinant Monoclonal Antibody Effector Functions by Fc N-Glycan Remodeling in Vitro," *Biotechnology Progress* 21 (2005): 1644–1652, <https://doi.org/10.1021/bp050228w>.
76. B. Peschke, C. W. Keller, P. Weber, I. Quast, and J. D. Lünemann, "Fc-Galactosylation of Human Immunoglobulin Gamma Isotypes Improves C1q Binding and Enhances Complement-Dependent Cytotoxicity," *Frontiers in Immunology* 8 (2017): 646–646, <https://doi.org/10.3389/fimmu.2017.00646>.
77. G. Ashwell and J. Harford, "Carbohydrate-Specific Receptors of the Liver," *Annual Review of Biochemistry* 51 (1982): 531–554, <https://doi.org/10.1146/annurev.bi.51.070182.002531>.
78. R. D. Gordon, P. Sivarajah, M. Satkunarajah, et al., "X-Ray Crystal Structures of Rabbit N-Acetylglucosaminyltransferase I (GnT I) in Complex With Donor Substrate Analogues," *Journal of Molecular Biology* 360, no. 1 (2006): 67–79, <https://doi.org/10.1016/j.jmb.2006.04.058>.
79. M. J. Gramer, J. J. Eckblad, R. Donahue, et al., "Modulation of Antibody Galactosylation Through Feeding of Uridine, Manganese Chloride, and Galactose," *Biotechnology and Bioengineering* 108, no. 7 (2011): 1591–1602, <https://doi.org/10.1002/bit.23075>.
80. M. M. St. Amand, D. Radhakrishnan, A. S. Robinson, and B. A. Ogunnaike, "Identification of Manipulated Variables for a Glycosylation Control Strategy," *Biotechnology and Bioengineering* 111, no. 10 (2014): 1957–1970, <https://doi.org/10.1002/bit.25251>.
81. E. Wells, L. Song, M. Greer, et al., "Media Supplementation for Targeted Manipulation of Monoclonal Antibody Galactosylation and Fucosylation," *Biotechnology and Bioengineering* 117, no. 11 (2020): 3310–3321, <https://doi.org/10.1002/bit.27496>.
82. D. C. Wong, N. S. Wong, J. S. Goh, L. M. May, and M. G. Yap, "Profiling of N-Glycosylation Gene Expression in CHO Cell Fed-Batch Cultures," *Biotechnology and Bioengineering* 107, no. 3 (2010): 516–528, <https://doi.org/10.1002/bit.22828>.
83. L. Meuris, F. Santens, G. Elson, et al., "GlycoDelete Engineering of Mammalian Cells Simplifies N-Glycosylation of Recombinant Proteins," *Nature Biotechnology* 32, no. 5 (2014): 485–489, <https://doi.org/10.1038/nbt.2885>.
84. R. K. Grainger and D. C. James, "CHO Cell Line Specific Prediction and Control of Recombinant Monoclonal Antibody N -Glycosylation," *Biotechnology and Bioengineering* 110, no. 11 (2013): 2970–2983, <https://doi.org/10.1002/bit.24959>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting information file 1: biot70143-sup-0001-

SuppMat.docx **Supporting information file 2:**

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